UTHealth Houston

Fully Funded Ph.D. Positions – UTHealth Houston, Spring & Fall 2027

The School of Public Health at The University of Texas Health Science Center at Houston (UTHealth Houston) is inviting applications for multiple fully funded Ph.D. positions (Graduate Research Assistantships) starting in Spring and Fall 2027. Positions include a competitive stipend, full tuition and fee coverage, and conference travel support.

Ph.D. students will work under the mentorship of Dr. Kaiming Bi, applying advanced modeling, data science, and artificial intelligence (AI) methods to address pressing public health challenges. Students will also collaborate with the Center for Health Care Data (CHCD), the Center for Spatial Temporal Modeling for Applications in Population Sciences (CSMAPS), and the Texas Epidemic Public Health Institute (TEPHI). Upon joining, students will work with Dr. Bi to co-develop a research agenda, focusing on one or more of the following areas:

(1) Modeling Infectious Disease Dynamics: Developing models to study population-level disease transmission and forecasting the prevalence of infectious diseases.

(2) Spatial and Patient-Level Data Integration: Using granular data to explore spatial heterogeneity across various public health topics, including infectious diseases, chronic diseases, and opioid overdoses.

(3) Drivers of Emerging Outbreaks and Seasonal Epidemics: Identifying the environmental, climatic, demographic, and behavioral drivers of emerging disease outbreaks, and understanding why seasonal epidemics (e.g., influenza, RSV, COVID-19) differ in timing, intensity, and transmission across seasons.

(4) Synthetic Health Administrative Data: Building privacy-preserving synthetic databases that mirror health administrative data, such as insurance claims, to enable secure data sharing and reproducible research.

(5) Methodological Innovation: Developing and applying dynamic (compartmental) transmission models, agent-based models, statistical models, machine learning models, and AI-based foundation models and large language models (LLMs).

Eligibility Requirements:

(1) A B.S. or M.S. degree in Engineering, Mathematics, Statistics, Data Science, Computer Science, or a closely related field (M.S. preferred).

(2) Strong quantitative skills in modeling, statistics, and data analytics (experience with Python, deep learning, digital twins, or LLMs is a plus).

(3) Interest in solving public health challenges, particularly in infectious diseases (prior experience recommended but not required).

(4) Self-motivation and a strong passion for research (a demonstrated track record of publications will be prioritized).

How to Apply:

Applications will be reviewed on a rolling basis until all positions are filled. Interested candidates should send their CV to Dr. Kaiming Bi at Kaiming.Bi@uth.tmc.edu. Dr. Bi will help candidates identify the most suitable Ph.D. program within the School of Public Health before they formally apply (see: SPH Ph.D. Programs).

About UTHealth Houston:

Founded in 1972, UTHealth Houston is a premier academic health center within the Texas Medical Center (TMC)—the largest medical complex in the world. TMC employs over 106,000 people, facilitates 10 million patient encounters annually, and is home to world-renowned institutions like Texas Children's Hospital and MD Anderson Cancer Center. UTHealth Houston offers a supportive, vibrant environment to launch your research career.